

Questions

MathonGo

Q1 - 2024 (04 Apr Shift 1)

A square is inscribed in the circle $x^2 + y^2 - 10x - 6y + 30 = 0$. One side of this square is parallel to $y = x + 3$. If (x_i, y_i) are the vertices of the square, then $\sum (x_i^2 + y_i^2)$ is equal to:

(1) 148

(2) 152

(3) 160

(4) 156

Q2 - 2024 (04 Apr Shift 2)

Let C be a circle with radius $\sqrt{10}$ units and centre at the origin. Let the line $x + y = 2$ intersects the circle C at the points P and Q . Let MN be a chord of C of length 2 unit and slope -1. Then, a distance (in units) between the chord PQ and the chord MN is

(1) $3 - \sqrt{2}$ (2) $\sqrt{2} + 1$ (3) $\sqrt{2} - 1$ (4) $2 - \sqrt{3}$

Q3 - 2024 (05 Apr Shift 1)

Let a circle C of radius 1 and closer to the origin be such that the lines passing through the point $(3, 2)$ and parallel to the coordinate axes touch it. Then the shortest distance of the circle C from the point $(5, 5)$ is :

(1) $2\sqrt{2}$ (2) $4\sqrt{2}$

(3) 4

(4) 5

Q4 - 2024 (05 Apr Shift 2)

Questions

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Let $ABCD$ and $AEFG$ be squares of side 4 and 2 units, respectively. The point E is on the line segment AB and the point F is on the diagonal AC . Then the radius r of the circle passing through the point F and touching the line segments BC and CD satisfies:

(1) $r = 0$

(2) $2r^2 - 4r + 1 = 0$

(3) $2r^2 - 8r + 7 = 0$

(4) $r^2 - 8r + 8 = 0$

Q5 - 2024 (05 Apr Shift 2)

Let the circle $C_1 : x^2 + y^2 - 2(x + y) + 1 = 0$ and C_2 be a circle having centre at $(-1, 0)$ and radius 2. If the line of the common chord of C_1 and C_2 intersects the y -axis at the point P , then the square of the distance of P from the centre of C_1 is :

(1) 2

(2) 1

(3) 4

(4) 6

Q6 - 2024 (06 Apr Shift 1)

A circle is inscribed in an equilateral triangle of side of length 12. If the area and perimeter of any square inscribed in this circle are m and n , respectively, then $m + n^2$ is equal to

(1) 408

(2) 414

(3) 396

(4) 312

Q7 - 2024 (06 Apr Shift 1)

Questions

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Let C be the circle of minimum area touching the parabola $y = 6 - x^2$ and the lines $y = \sqrt{3}|x|$. Then, which one of the following points lies on the circle C ?

- (1) (1, 2)
- (2) (1, 1)
- (3) (2, 2)
- (4) (2, 4)

Q8 - 2024 (08 Apr Shift 1)

Let the circles $C_1 : (x - \alpha)^2 + (y - \beta)^2 = r_1^2$ and $C_2 : (x - 8)^2 + \left(y - \frac{15}{2}\right)^2 = r_2^2$ touch each other externally at the point (6, 6). If the point (6, 6) divides the line segment joining the centres of the circles C_1 and C_2 internally in the ratio 2 : 1, then $(\alpha + \beta) + 4(r_1^2 + r_2^2)$ equals

- (1) 125
- (2) 130
- (3) 110
- (4) 145

Q9 - 2024 (09 Apr Shift 1)

Let a circle passing through (2, 0) have its centre at the point (h, k) . Let (x_c, y_c) be the point of intersection of the lines $3x + 5y = 1$ and $(2 + c)x + 5c^2y = 1$. If $h = \lim_{c \rightarrow 1} x_c$ and $k = \lim_{c \rightarrow 1} y_c$, then the equation of the circle is :

- (1) $25x^2 + 25y^2 - 2x + 2y - 60 = 0$
- (2) $5x^2 + 5y^2 - 4x + 2y - 12 = 0$
- (3) $5x^2 + 5y^2 - 4x - 2y - 12 = 0$
- (4) $25x^2 + 25y^2 - 20x + 2y - 60 = 0$

Q10 - 2024 (09 Apr Shift 1)

Questions

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Let the centre of a circle, passing through the points $(0, 0)$, $(1, 0)$ and touching the circle $x^2 + y^2 = 9$, be (h, k) . Then for all possible values of the coordinates of the centre (h, k) , $4(h^2 + k^2)$ is equal to _____.

Questions

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Answer Key

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Q2 (1) mathongo

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Q3 (3) mathongo

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Q4 (4) mathongo

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Q5 (1) mathongo

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Q6 (1) mathongo

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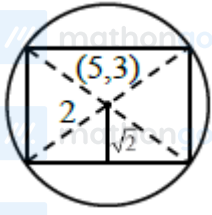
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#MathBoleTohMathonGo

Solutions

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Q1



$$y = x + c \quad \&$$

$$y = x + c \quad \& \quad x + y + d = 0$$

$$\left| \frac{5-3+c}{\sqrt{2}} \right| = \sqrt{2} \quad \left| \frac{8+d}{\sqrt{2}} \right| = \sqrt{2}$$

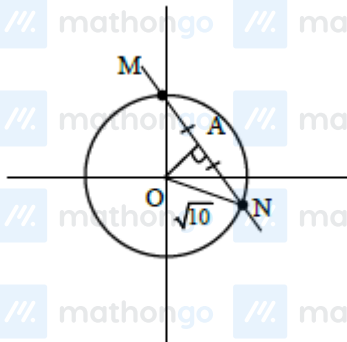
$$|c+2| = 2 \quad 8+d = \pm 2$$

$$c = 0, -4 \quad d = -10, -6$$

$$\text{pts } (5, 5), (3, 3), (7, 3), (5, 1)$$

$$\sum (x_i^2 + y_i^2) = 25 + 25 + 9 + 9 + 49 + 9 + 25 + 1 = 152$$

Q2



$$C: x^2 + y^2 = 10$$

$$AN = \frac{MN}{2} = 1$$

$$\therefore \text{In } \triangle OAN \rightarrow (ON)^2 = (OA)^2 + (AN)^2$$

$$10 = (OA)^2 + 1 \rightarrow OA = 3$$

Perpendicular distance of center from

$$PQ = \frac{|0+0-2|}{\sqrt{2}} = \sqrt{2}$$

Perpendicular distance between MN and

$$PQ = OA + \sqrt{2} \text{ or } |OA - \sqrt{2}|$$

$$= 3 + \sqrt{2} \text{ or } 3 - \sqrt{2}$$

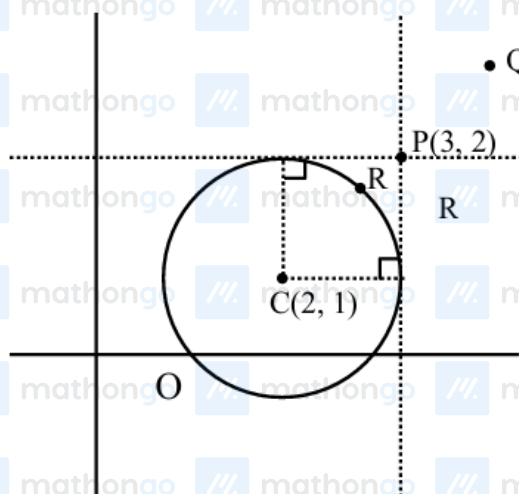
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Solutions

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Q3



Coordinates of the centre will be $(2, 1)$

Equation of circle will be

$$(x - 2)^2 + (y - 1)^2 = 1$$

$$QC = \sqrt{(5 - 2)^2 + (5 - 1)^2}$$

$$QC = 5$$

shortest distance

$$= RQ = CQ - CR$$

$$= 5 - 1$$

$$= 4$$

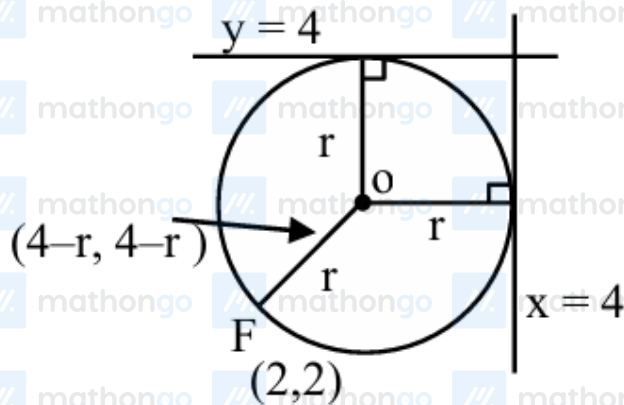
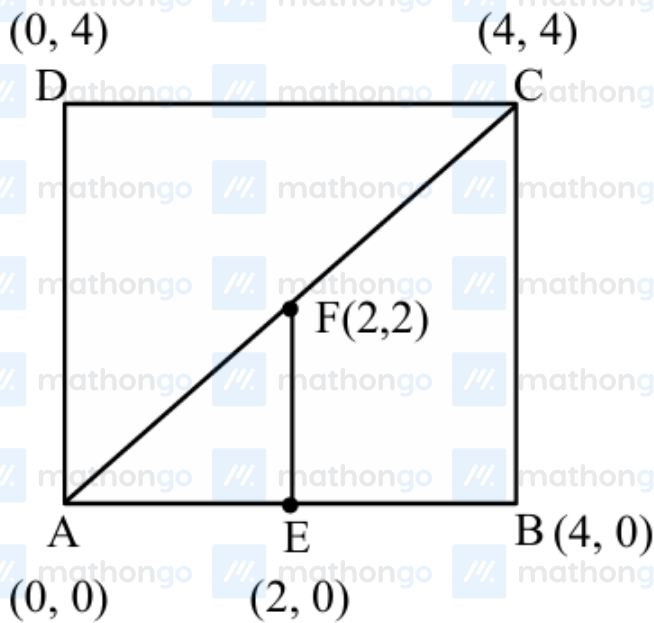
Q4

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Solutions

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$$OF^2 = r^2$$

$$(2-r)^2 + (2-r)^2 = r^2$$

$$r^2 - 8r + 8 = 0$$

Q5

$$S_1 : x^2 + y^2 - 2x - 2y + 1 = 0$$

$$S_2 : x^2 + y^2 + 2x - 3 = 0$$

$$\text{Common chord} = S_1 - S_2 = 0$$

$$-4x - 2y + 4 = 0$$

$$2x + y = 2 \Rightarrow P(0,2)$$

$$d_{(c,p)}^2 = (1-0)^2 + (2-1)^2 = 2$$

Q6

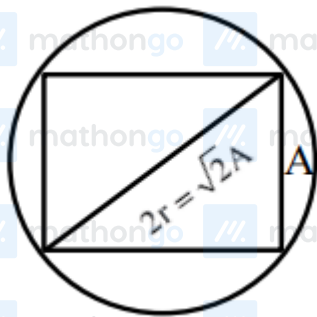
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Solutions

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$$\therefore r = \frac{\Delta}{s} = \frac{\sqrt{3}a^2}{4 \cdot \frac{3a}{2}} = \frac{a}{2\sqrt{3}} = \frac{12}{2\sqrt{3}} = 2\sqrt{3}$$



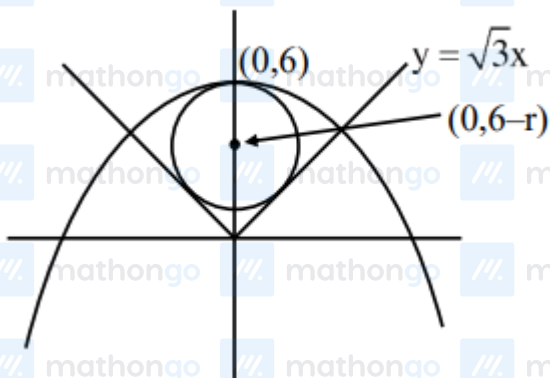
$$\therefore A = r\sqrt{2} = 2\sqrt{6}$$

$$\text{Area} = m = A^2 = 24$$

$$\text{Perimeter} = n = 4A = 8\sqrt{6}$$

$$\therefore m + n^2 = 24 + 384 = 408$$

Q7



Equation of circle

$$x^2 + (y - (6 - r))^2 = r^2$$

$$\text{touches } \sqrt{3}x - y = 0$$

$$p = r$$

$$\frac{|0 - (6 - r)|}{2} = r$$

$$|r - 6| = 2r$$

$$r = 2$$

$$\therefore \text{Circle } x^2 + (y - 4)^2 = 4$$

(2, 4) Satisfies this equation

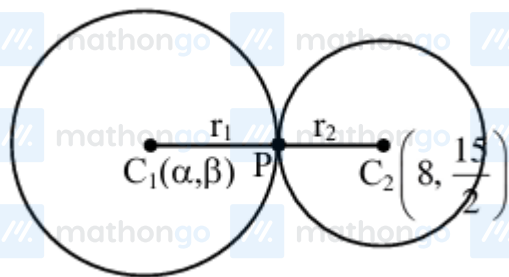
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Solutions

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Q8



$$\begin{array}{c} 2 : 1 \\ (\alpha, \beta) \quad C_1 \quad P \quad C_2 \quad \left(8, \frac{15}{2}\right) \\ (6, 6) \end{array}$$

$$\therefore \frac{16 + \alpha}{3} = 6 \text{ and } \frac{15 + \beta}{3} = 6$$

$$\Rightarrow (\alpha, \beta) \equiv (2, 3)$$

$$\text{Also, } C_1 C_2 = r_1 + r_2$$

$$\Rightarrow \sqrt{(2-8)^2 + \left(3 - \frac{15}{2}\right)^2} = 2r_2 + r_2$$

$$\Rightarrow r_2 = \frac{5}{2} \Rightarrow r_1 = 2r_2 = 5$$

$$\begin{aligned} \therefore (\alpha + \beta) + 4(r_1^2 + r_2^2) \\ = 5 + 4\left(\frac{25}{4} + 25\right) = 130 \end{aligned}$$

Q9

Solutions

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$$(2+c)x + 5c^2 \left(\frac{1-3x}{5} \right) = 1$$

$$x = \frac{1-c^2}{2+c-3c^2}, y = \frac{1-3x}{5} = \frac{c-1}{5(2+c-3c^2)}$$

$$h = \lim_{c \rightarrow 1} \frac{(1-c)(1+c)}{(1-c)(2+3c)} = \frac{2}{5}$$

$$K = \lim_{c \rightarrow 1} \frac{c-1}{-5(c-1)(3c+2)} = -\frac{1}{25}$$

$$\text{Centre} \left(\frac{2}{25}, -\frac{1}{25} \right)$$

$$r = \sqrt{\left(2 - \frac{2}{5} \right)^2 + \left(0 - \frac{1}{25} \right)^2} = \sqrt{\frac{64}{25} + \frac{1}{625}}$$

$$r = \frac{\sqrt{161}}{25}$$

$$\left(x - \frac{2}{5} \right)^2 + \left(y + \frac{1}{25} \right)^2 = \frac{161}{125}$$

$$\Rightarrow 25x^2 + 25y^2 - 20x + 2y - 60 = 0$$

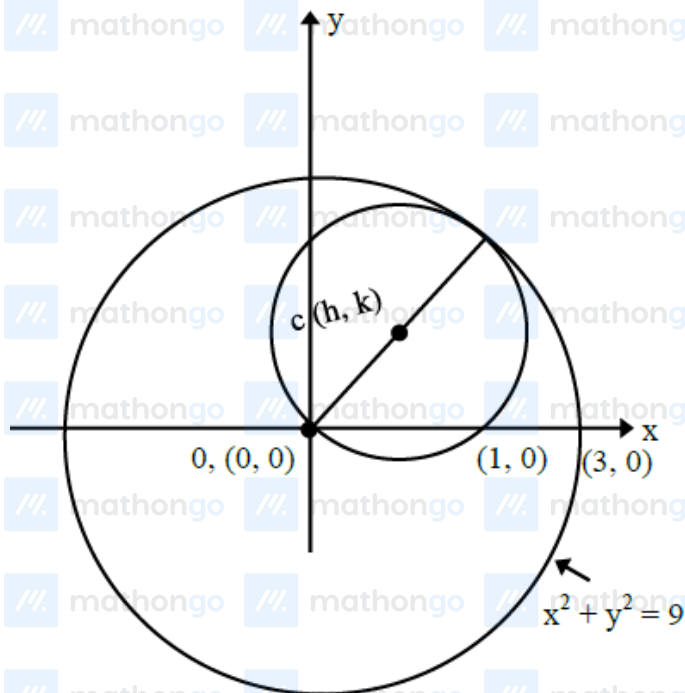
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Solutions

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$$(x - h)^2 + (y - k)^2 = h^2 + k^2$$

$$x^2 + y^2 - 2hx - 2ky = 0$$

$\therefore$ passes through $(1, 0)$

$$\Rightarrow 1 + 0 - 2h = 0$$

$$\Rightarrow h = 1/2$$

$$\therefore OC = \frac{OP}{2}$$

$$\sqrt{\left(\frac{1}{2}\right)^2 + k^2} = \frac{3}{2}$$

$$\frac{1}{4} + k^2 = \frac{9}{4}$$

$$k^2 = 2$$

$$k = \pm\sqrt{2}$$

$\therefore$ Possible coordinate of

$$C(h, k) \left(\frac{1}{2}, \sqrt{2} \right) \left(\frac{1}{2}, -\sqrt{2} \right)$$

$$4(h^2 + k^2) = 4\left(\frac{1}{4} + 2\right) = 4\left(\frac{9}{4}\right) = 9$$